

SCHOOL OF COMPUTING (SEM II – 2024/2025)

SCST1223 – PSDA

Exercise (Chapter 5 – Part1)

QUESTION 1

In a study of 31.95 million for a country's population, 533 thousand Covid-19 cases have been detected within 15 months. Construct a 95% confidence interval for the proportion of people that confirms Covid-19. Is there a significant difference between people who confirms with Covid-19 and people who do not infect?

QUESTION 2

- (a) In a study of women's 100-meter performance of 25 university students, mean time is 19.5 seconds with standard deviation of 9.88 seconds were recorded. Construct a 90% confidence interval to estimate the mean.
- (b) In a random sample of 360 food stores, it was found that 160 of these food stores offers special promotions. Construct a 99% confidence interval to estimate the population of food stores that offers promotion.

QUESTION 3

Mendelian genetics conducted his famous genetics experiments with peas, one sample of offspring consisted of 428 green peas and 152 yellow peas.

- (i) Compute a point estimate of yellow peas. (2 marks)
- (ii) Construct and interpret a 95% confidence interval estimate of the percentage of yellow peas. (5 marks)
- (iii) Based on his theory of genetics, Mendel expected that 25% of the offspring peas would be yellow. Given that the percentage of offspring yellow peas is not 25%, do the results contradict Mendel's theory? Justify your answer. (3 marks)

QUESTION 4

- a) In the February 20th issue of the East Daily Times, it was reported that $n = 418$ insured patients were surveyed on their opinions about insurance rates. Of the 418 surveyed, $Y = 280$ blamed rising insurance rates on large court settlements against doctors. With 95% confidence, estimate the parameter p , that is, the proportion of all insured patients who blame rising insurance rates on large court settlements against doctors.
- b) A random sample of 126 police officers subjected to constant inhalation of automobile exhaust fumes in downtown Cairo, had an average blood lead level concentration of 29.2 $\mu\text{g/dl}$. Assume x , the blood lead level of a randomly selected policeman, is normally distributed with a standard deviation of $\sigma = 7.5 \mu\text{g/dl}$. Historically, it is known that the average blood lead level concentration of humans with no exposure to automobile exhaust is 18.2 $\mu\text{g/dl}$. Construct 90% confidence interval estimator for average blood lead level.
- (5 marks)

QUESTION 5

In a random sample of 350 food shops, it was found that 130 of them had special offers.

- a) Calculate an approximate 95% confidence interval for the population of all food shops with special offers. (4 marks)
- b) Estimate the new size of a random sample required for an approximate 95% confidence interval for the proportion calculated in (a) that has margin of error of 0.04. (4 marks)

QUESTION 6

A survey has been conducted among pet's shop in Malaysia. From 350 random samples collected, there are 130 pet's shop involved in Year End Sale (YES).

- a) Calculate an approximate 95% confidence interval for the population of all pet's shops that join YES. (6 marks)
- b) Estimate the new size of a random sample required for an approximate 95% confidence interval for the proportion calculated in (a) that has margin of error of 0.04. (4 marks)